Talks at the JMM

Terence Tao · 17 March 2024

https://terrytao.wordpress.com/2024/03/17/talks-at-the-jmm/

tao2typ · converted to Typst 2026-10-14 (UTC)

Earlier this year, I gave a series of lectures at the Joint Mathematics Meetings at San Francisco. I am uploading here the slides for these talks:

- "Machine assisted proof" (Video here)
- "Translational tilings of Euclidean space" (Video here)
- "Correlations of multiplicative functions" (Video here)

I also have written a text version of the first talk, which has been submitted to the Notices of the American Mathematical Society.

Comments (11)

Anonymous · 18 March 2024

first!

Anonymous · 18 March 2024

Thanks for posting the Slides and Videos for your talks. A role model for other mathematicians.

Anonymous · 18 March 2024

https://mathoverflow.net/questions/467247/subgroup-of-p-adic-units

> **Anonymous** · 19 March 2024
>
> Thanks for sharing!

Anonymous · 19 March 2024

Corrections to AMS article (in order):
“use computers assist” -> “use computers to assist”
“recent proof of Gowers, Green, Manners and myself” -> [GGMT23] is not referenced, so add it here
“an superintelligent” -> “a superintelligent”
“The computation could produce incur numerical errors” -> delete “produce”
“each step in a maathematical proof” -> “... mathematical ...”
“..“only“ eleven years..” -> bad double quotes
“measured to the influence” -> “measured the influence”
“results on other PDEs” -> “results of other PDEs”
“ubiquitious” -> “ubiquitous”
“scaleable” -> “scalable”
[Thanks, this will be incorporated into the next revision of the ms. -T]

Anonymous · 26 March 2024

Some possible errors in the AMS article:
“Mathematicians have relied on upon computers” – should possibly be “Mathematicians have relied on computers” or “Mathematicians have relied upon computers”.
“Theorem 0.1 (Boolean Pythaogrean triples” – should probably be “Theorem 0.1 (Boolean Pythagorean triples”.
“The computation could produce incur numerical errors” – should possibly be “The computation could produce numerical errors” or “The computation could incur numerical errors”.
[Thanks, these will be corrected in the final version of the ms -T]

Anonymous · 26 March 2024

How are the logarithmically averaged Chowla conjecture for $k = 1$ and the logarithmically averaged prime number theorem ($\sum_{nlex} \frac{\Lambda(n)}{n} = \log(x) - \gamma + o(1)$) related?
Are they equivalent? If not, what is the difference in light of the normal Chowla conjecture being equivalent to the normal prime number theorem?
Why doesn’t the logarithmically averaged prime number theorem have a similarly simple proof?

> 
> **Terence Tao** · 2 April 2024
>
> The closest analogues here are the elementary bound $\sum_{n \leq x} \mu(n)/n = O(1)$ and Mertens' theorem $\sum_{n \leq x} \Lambda(n)/n = \log x + O(1)$, which are proven the same way: one computes

$\sum_{n\le x}(\sum_{d\mid n} f(d)) = \sum_{d\le x} f(d)\sum_{m\le x/d} 1$ in two different ways for $f=\mu$ and for $f=\Lambda$. I think of Mertens' theorem as the natural logarithmically averaged version of the PNT. The stronger asymptotic you mention is analogous to the strengthened k=1 Chowla bound $\sum_{n\le x}\mu(n)/n = o(1)$, which is elementarily equivalent to PNT. (Note by the way that Chowla only claims the far weaker bound $o(\log x)$ here.)

 **Jas, the Physicist** · 26 March 2024

I enjoyed the "machine assisted proof" talk. Thank you.

 **Anonymous** · 27 March 2024

It was so great to meet you! Hope everything's going well :)

 **Anonymous** · 28 March 2024

What do you think of Kirti Joshi's proof of the abc conjecture?